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Deep Learning - Based Brain Tumor Detection from MRI Images using Transfer Learning and Advanced Optimization Techniques.

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1.Introduction

Brain tumors are abnormal tissue growths within the brain that disrupt normal neurological functions and, if left untreated, can result in severe complications or death. Accurate and early diagnosis plays a crucial role in improving patient outcomes. Among the various types, glioma, meningioma, and pituitary tumors are the most commonly diagnosed. Traditional diagnosis via manual interpretation of MRI scans, while widely practiced, is time-intensive and prone to inter-observer variability and human fatigue (Kaur et al., 2023; Lata et al., 2024).

MRI remains the gold standard for non-invasive brain tumor imaging due to its high-resolution and contrast sensitivity. However, with the exponential growth of medical imaging data, automated classification systems powered by Deep Learning (DL) have gained momentum. These systems offer faster and more consistent decision support, especially when trained on diverse and balanced datasets (Bharali et al., 2024). Convolutional Neural Networks (CNNs), a powerful subset of DL, are particularly effective at learning spatial hierarchies from image data, making them ideal for tumor classification (Castelino et al., 2024).

In this project, we designed and implemented a custom CNN model from scratch to classify brain MRI images into four categories: glioma, meningioma, pituitary tumor, and no tumor. To combat class imbalance - a common challenge in medical datasets. We applied random oversampling, and used extensive data augmentation techniques (e.g., flipping, shifting, zooming, and rotation) to improve generalization. To prevent overfitting during training, incorporated dropout layers and trained the model over 100 epochs, achieving a test accuracy of 92% and precision 93% F1-scores of 0.92 and recall 0.91 for the no tumor and pituitary classes, respectively.

Building on the strong performance of the custom-built CNN model, the next phase will focus on integrating transfer learning using pre-trained architectures such as EfficientNet, ResNet, and Xception to enhance classification accuracy and reduce training time. Additionally, we aim to explore ensemble learning strategies, which combine the strengths of multiple models to improve prediction stability and clinical reliability (Shah et al., 2023; Novsheena et al., 2025). These enhancements aim to develop a robust, scalable diagnostic system capable of assisting radiologists in accurately identifying brain tumors in real-world clinical environments.

1.1 Background

Brain tumors are abnormal growths that develop within brain tissues and can be classified as benign or malignant. Among the common types, gliomas originate from glial cells and are often aggressive, with glioblastoma being the most lethal form. Meningiomas arise from the meninges, the protective layers of the brain, and are typically slow-growing and noncancerous. Pituitary tumors develop in the pituitary gland and may affect hormone levels, causing systemic health issues. Treatment approaches vary by tumor type but commonly involve surgery, radiation therapy, or chemotherapy (Louis et al., 2016;Ostrom et al., 2022).

The exact causes of brain tumors remain uncertain; however, several risk factors have been identified. These include genetic mutations, prolonged exposure to ionizing radiation, and certain inherited syndromes such as Li-Fraumeni and neurofibromatosis. In the case of pituitary tumors,

hormone dysregulation may play a role. Treatment selection depends on tumor size, type, location, and the patient's overall health. Surgical removal is often the first-line approach, while radiation or drug therapy may follow for residual or inoperable tumors (Stupp et al., 2005; Ostrom et al., 2022).

In recent years, deep learning has revolutionized medical image analysis by enabling systems to automatically learn complex features from imaging data. In particular, Convolutional Neural Networks (CNNs) are highly effective for processing visual inputs like MRI scans, making them suitable for tumor classification tasks. CNNs work by learning spatial hierarchies in images, allowing them to detect patterns specific to different tumor types. This project leverages CNNs to classify brain MRI images into glioma, meningioma, pituitary tumor, or no tumor, enhancing diagnostic speed and reliability (Litjens et al., 2017; Afshar et al., 2020).

1.2 Research Questions

- How do data augmentation, class balancing, and regularization techniques influence the generalization ability, accuracy, and clinical relevance of CNN-based models for multiclass brain tumor classification using MRI scans?
- How can a custom-built Convolutional Neural Network (CNN) be designed and optimized to accurately classify brain tumors from MRI images into multiple categories: glioma, meningioma, pituitary tumor, and no tumor?
- What is the comparative performance of a handcrafted CNN model versus advanced transfer learning architectures such as EfficientNet, ResNet, and Xception in improving diagnostic precision and clinical applicability for brain tumor detection?

1.3 AIM

To develop an accurate and efficient deep learning framework for multiclass brain tumor classification using MRI images, leveraging both custom CNNs and transfer learning. The system aims to address class imbalance, overfitting, and limited data to support clinical diagnostics.

1.4 Objectives

- To collect and preprocess a labelled MRI dataset with four classes: glioma, meningioma, pituitary tumor, and normal.
- To apply image normalization, resizing, and class balancing to address dataset limitations and ensure fair model training.
- To implement and fine-tune multiple transfer learning-based CNN architectures (EfficientNet, ResNet, Xception) for multiclass classification.
- To use data augmentation methods (e.g., rotation, flipping, brightness change) to enhance dataset diversity and mitigate overfitting.
- To evaluate and compare the model performances using accuracy, precision, recall, F1-score, and confusion matrices.

1.5 Literature Review

Gunjan Sharma et al. (2024) developed a CNN-based model for binary classification of brain tumors using the BT Image Dataset. The study involved preprocessing steps like resizing MRI images to 250×250 pixels, normalization, and applying augmentation techniques such as rotation (up to 30°), brightness adjustment, and horizontal/vertical flipping to improve model generalization. The dataset was split in an 8:1:1 ratio for training, validation, and testing. Their CNN model achieved a classification accuracy of 97.29%, with precision and recall of 98% and 97.85%, respectively. This approach effectively demonstrated the role of CNNs in early brain tumor detection and offered a robust method for clinical application.

Oussama Bouguerra et al. (2022) proposed a transfer learning-based approach for binary classification of brain tumors using MRI images. They employed VGG16 and VGG19 architectures with enhanced preprocessing (cropping, resizing to 224×224), image enhancement (CLAHE, histogram equalization), and data augmentation, increasing the tumor and normal images to 1,238 and 980, respectively. The dataset was split 70% for training, 10% validation, and 20% testing. Their method achieved 99.77% accuracy, showing that transfer learning with minimal data and effective augmentation significantly boosts diagnostic performance in resource-constrained settings.

Sahar Gull et al. (2021) developed a CNN-based unified framework for fully automated segmentation and classification of brain tumors using 3D-MRI scans from the BRATS 2018 dataset. Their method includes preprocessing, augmentation, and segmentation followed by binary classification using Dense-Net and Dark-Net classifiers. The framework achieved a segmentation DSC of 97.91% and accuracy of 98.67%. For classification, Dense-Net outperformed Dark-Net with 98.14% DSC and 98.26% accuracy, indicating its superior reliability. This approach demonstrates promising potential for clinical implementation in early and accurate tumor detection.

Talukder et al. (2023) proposed a deep learning-based brain tumor classification method using transfer learning to enhance diagnostic accuracy. The study utilized the Figshare MRI brain tumor dataset with 3,064 images and applied models like Xception, ResNet50V2, InceptionResNetV2, and DenseNet201. Extensive preprocessing and model fine-tuning were performed. ResNet50V2 achieved the highest classification accuracy of 99.68%, outperforming other models. This approach offers a fast, reliable tool to assist clinicians in early and accurate tumor detection.

Mittal et al. (2024) proposed a deep learning-based approach for the early and multiclass classification of brain tumors using the DenseNet201 model. The study utilized a dataset of 7,023 brain MRI scans categorized into glioma, meningioma, pituitary tumor, and no tumor. Preprocessing involved image augmentation with ImageDataGenerator. The DenseNet201 model achieved 99% accuracy, along with high precision, recall, and F1-score, proving its reliability. This research highlights the critical role of prompt tumor detection in improving treatment planning and patient outcomes.

1.5.1 Analysis of Literature

The reviewed literature highlights significant advancements in brain tumor detection using deep learning, particularly CNNs and transfer learning models. Studies by Sharma et al., (2024) Bouguerra et al., (2022) and Gull et al., (2021) focus on binary classification and segmentation, showcasing the strength of CNNs and VGG-based architectures in handling limited datasets with strong performance through data augmentation and preprocessing. Other works, like those by Talukder et al., (2023) and Mittal et al., (2024) emphasize multi-class classification using transfer learning models such as ResNet50V2 and DenseNet201, achieving exceptional accuracy (up to 99.68%) on large MRI datasets. Overall, these approaches demonstrate the effectiveness of DL models in automating and enhancing brain tumor diagnosis, yet there remains a need for unified models combining segmentation and classification, deeper exploration of custom architectures, and robust testing in real-world clinical settings.

1.5.2 Research Gap

Most existing studies rely heavily on pre-trained models like DenseNet, ResNet, and VGG without customizing architectures for domain-specific features in brain MRI data. Few have explored designing CNNs from scratch, limiting innovation in architecture tailored for multiclass classification tasks. Your work fills this gap by developing a custom CNN optimized specifically for glioma, meningioma, pituitary, and no-tumor classes.

Another critical gap is the handling of class imbalance, which many works overlook, resulting in biased predictions and poor minority class performance. Your study addresses this with random oversampling and comprehensive augmentation techniques, ensuring balanced learning across all tumor categories, as seen in strong F1-scores for underrepresented classes.

Furthermore, current models rarely combine custom CNNs with transfer learning or ensemble strategies to boost performance and generalization. Your phased approach - starting with a custom-built model, followed by transfer learning integration and ensemble exploration - creates a novel, hybrid pipeline aimed at producing a clinically reliable and scalable diagnostic tool.

2. Summary of progress to date

This section presents the progress made so far in developing a brain tumor classification model using MRI images. It covers dataset acquisition, preprocessing, exploratory analysis,class balancing, data augmentation, model architecture, training outcomes, and evaluation.

2.1 Dataset Acquisition and Exploratory Data Analysis

The dataset used in this study was obtained from Kaggle's Brain Tumor MRI Dataset. It contains T1-weighted contrast-enhanced MRI scans of human brains categorized into four classes: glioma, meningioma, pituitary tumor, and no tumor. These images are available in PNG format and were resized to 130×130 pixels during preprocessing to ensure uniformity and reduce computational overhead.

The dataset is split into two subsets: training and testing. The training set comprises 1,457 pituitary, 1,321 glioma, 1,595 no tumor, and 1,339 meningioma images. The test set contains 300 images each of glioma and pituitary tumors, 405 no tumor images, and 306 meningioma images. While the distribution is relatively balanced, the no tumor class contains slightly more samples in both subsets.Figure 1: presents sample images from each category, while Figure 2: illustrates the distribution of images across the different classes in both training and testing sets.

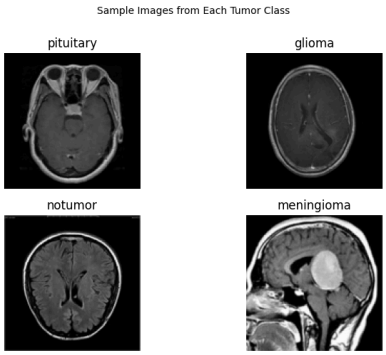


Figure - 1: Sample images

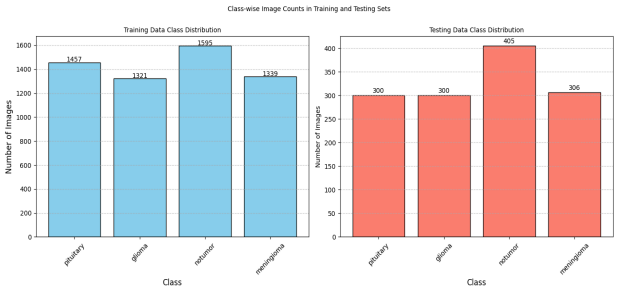


Figure - 2 : Each class data distribution

2.2 Preprocessing and Augmentation

All MRI images are resized to 130×130 pixels and converted to RGB for consistent input shape. Data augmentation like rotation, zoom, and flipping is applied to enhance variability and reduce overfitting.Images are normalized (pixel values scaled between 0 and 1) to improve training stability.

To address class imbalance, augmented images are generated more for underrepresented classes to equalize training counts using random sampling. Figure 3: shows examples of augmented images, and Figure 4: compares class distribution before and after balancing.

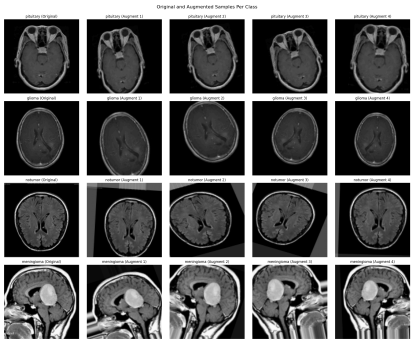


Figure -3 : Augmented Images

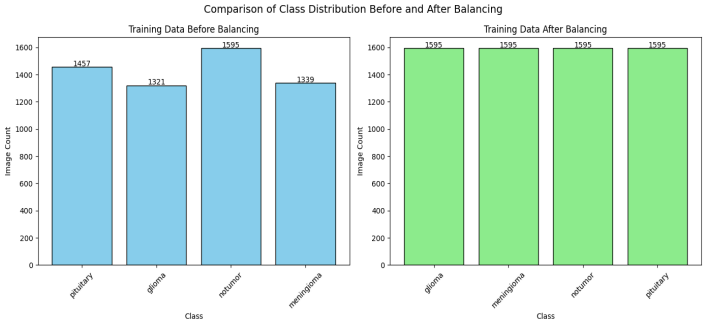


Figure - 4 : Before and After class balancing data distribution.

2.3 Model Development and Training

To train an effective brain tumor classifier, we began by splitting the balanced dataset into training and validation sets using an 80-20 stratified approach to maintain class distribution. We designed a Convolutional Neural Network (CNN) architecture consisting of three convolutional layers with increasing filter sizes (32, 64, 128), each followed by max-pooling to progressively reduce spatial dimensions. A fully connected dense layer with 512 units and a dropout layer (rate = 0.5) were added to mitigate overfitting, ending with a softmax layer for multiclass prediction.

For preprocessing, we employed image augmentation techniques such as rotation, zoom, shifts, shear, and horizontal flipping using ImageDataGenerator to enhance model generalization. All pixel values were rescaled to normalize input. The model was compiled with the Adam optimizer and trained using categorical cross-entropy loss over 100 epochs, with performance tracked on the validation set. This setup allowed the model to effectively capture spatial patterns critical for distinguishing between tumor types. Figure 5: illustrates the training and validation accuracy and loss trends over epochs.

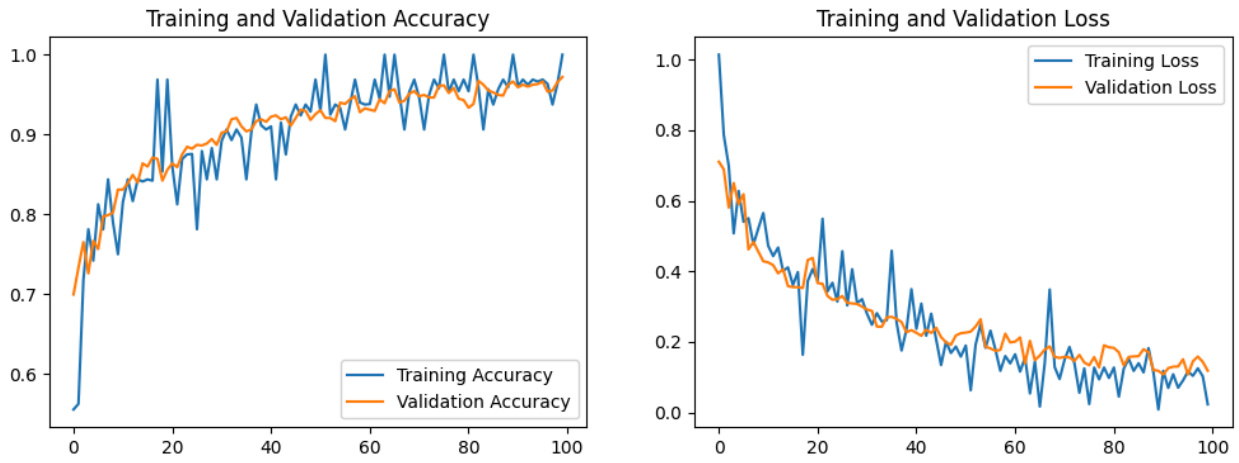


Figure - 5 : Training and validation Accuracy Loss Plot

2.4 Evaluation and Results

We evaluated the trained model on the unseen test dataset using accuracy, precision, recall, and F1-score. A classification report and confusion matrix were generated to analyze performance across all four tumor classes. The model achieved strong accuracy overall, with minor confusion between some classes. These metrics confirmed the model's reliability for brain tumor classification. Table 1: summarizes the key performance metrics, while Figure 6: presents the confusion matrix, showing only minor misclassifications between some tumor classes.

Model	Accuracy	Precision	Recall	F1-score
CNN	0.92	0.93	0.91	0.92

Table -1 CNN performance results

Confusion Matrix

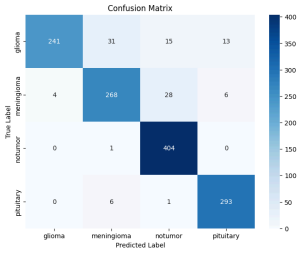


Figure - 6 Confusion Matrix

2.5 Tools and Libraries used:

- Google Colab - Cloud-based notebook environment
- Python 3 - Programming language used for implementation
- TensorFlow / Keras - For building, training, and evaluating the CNN model
- scikit-learn - For train-test split and performance metrics (accuracy, precision, recall, F1-score)
- Pandas & NumPy - For data handling and preprocessing
- Matplotlib & Seaborn - For visualizing data distributions, training history, and confusion matrix
- PIL (Python Imaging Library) - For image loading and resizing
- ImageDataGenerator - For image augmentation and normalization

3. Ethical, Legal, Professional and Social Issues

The current project focuses on building a convolutional neural network (CNN) to classify brain tumors from MRI images using a publicly available dataset. Since this dataset is anonymised and pre-curated for research use (Kaggle, 2022), formal ethics approval was not required. No patient-identifiable information is processed, and the project is conducted solely in a non-clinical, academic context. However, if the system were scaled for clinical use, ethics approval from a healthcare institution's review board would be mandatory to safeguard human rights and medical standards (Loftus and Wang, 2023).

Ethical Considerations

From an ethical standpoint, the project aligns with key biomedical principles - beneficence, non-maleficence, and justice (Beauchamp and Childress, 2019). The model is designed to support early diagnosis, which may assist radiologists in reducing manual error and expediting treatment. However, ethical risks arise in the form of dataset bias. For instance, if the dataset underrepresents specific demographics or tumor types, the model's predictions could be skewed, potentially disadvantageous to certain patient groups (Mehrabi et al., 2021). Moreover, the explainability of AI in medical settings is crucial to avoid over-reliance on algorithmic decisions by clinicians. Ethical deployment should always maintain a human-in-the-loop approach to uphold responsibility (Finlayson et al., 2021).

Legal Considerations

The project currently adheres to data protection standards, including the EU General Data Protection Regulation (GDPR), as no identifiable health data is collected or stored (European Union, 2016). If developed into a clinical application, legal issues will become more complex - covering areas such as patient consent, data minimization, algorithmic transparency, and medical

device compliance under MDR (EU Medical Device Regulation). Furthermore, any use of third-party datasets must respect licensing terms and intellectual property rights, particularly if the model is commercialised or integrated into clinical decision-making (Grote and Berens, 2020).

Professional Considerations

Professionally, the project follows responsible AI development practices such as transparency in data preprocessing, modular code design, and reproducibility. These standards support accountability and are consistent with IEEE's Ethically Aligned Design guidelines (IEEE, 2022). If extended into healthcare delivery, the model would require professional validation, including clinical trials and compliance with ISO standards like ISO/IEC 27001 for data security and ISO 13485 for medical device quality management (ISO, 2023). In future deployments, interdisciplinary collaboration with medical experts, AI ethicists, and legal advisors would be necessary.

Social Considerations

Socially, AI-based healthcare tools have transformative potential—but also raise concerns. A key issue is healthcare equity. If access to AI diagnostic tools is limited to certain regions or institutions, it could exacerbate existing inequalities in medical care (Vyas et al., 2022). In addition, public trust in AI must be earned through transparency, accessibility, and clear accountability mechanisms. There is also the concern of automation bias, where clinicians might defer excessively to machine-generated outputs (Rajkumar et al., 2022). The model, while promising, must be implemented with safeguards to ensure that its social benefits outweigh its risks.

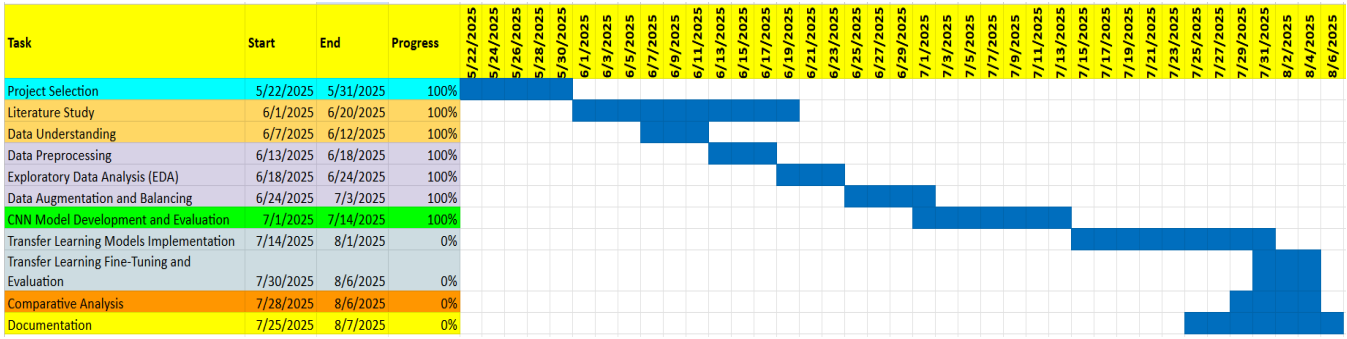
4. Project Plan

Project Management Approach

We adopted a hybrid project management strategy inspired by Agile principles to allow flexibility in experimentation while maintaining focus and accountability. Weekly iterations, structured tasks, and defined milestones ensured consistent progress. This approach enabled incremental model development, early-stage validations, and the ability to pivot based on results. Key project activities were aligned with both academic requirements and professional practices, simulating an industry-like development cycle..

Key Phases and Timeline Summary

The project progressed through weekly phases, starting with problem definition, domain research, and literature review. Mid-phase focused on data preprocessing, augmentation, and class balancing. The final phase involved implementing CNN and transfer learning models (next phase) , followed by evaluation and report writing. Figure 7: illustrates the overall timeline using a Gantt chart.



Task	Start	End	Progress
Project Selection	5/22/2025	5/31/2025	100%
Literature Study	6/1/2025	6/20/2025	100%
Data Understanding	6/7/2025	6/12/2025	100%
Data Preprocessing	6/13/2025	6/18/2025	100%
Exploratory Data Analysis (EDA)	6/18/2025	6/24/2025	100%
Data Augmentation and Balancing	6/24/2025	7/3/2025	100%
CNN Model Development and Evaluation	7/1/2025	7/14/2025	100%
Transfer Learning Models Implementation	7/14/2025	8/1/2025	0%
Transfer Learning Fine-Tuning and Evaluation	7/30/2025	8/6/2025	0%
Comparative Analysis	7/28/2025	8/6/2025	0%
Documentation	7/25/2025	8/7/2025	0%

Figure - 7 : Gantt Chart

Next phase

In the next phase of the project, we will focus on implementing advanced transfer learning techniques using state-of-the-art convolutional neural networks such as ResNet50, EfficientNetB0, and Xception. These models, pre-trained on large datasets like ImageNet, have proven effective in medical imaging tasks by enabling strong feature extraction even with limited labeled data . We plan to fine-tune selected layers of these networks by gradually unfreezing their top layers and training them with a lower learning rate to adapt to the nuances of brain tumor MRI images. This approach is expected to enhance classification accuracy, particularly in differentiating visually similar tumor types.

Beyond individual models, we will also explore ensemble techniques like soft voting to improve model robustness and reduce bias (Logan et al., 2021). This enhances prediction stability - critical in medical diagnostics. we will also document model configurations to support reproducibility and potential clinical deployment.

Challenges Encountered

One of the most significant challenges was limited GPU computing resources when using free-tier platforms like Google Colab. Deep learning models such as CNN and transfer learning models require high memory bandwidth and long training cycles, which strained runtime limits. This limitation constrained the extent of hyperparameter tuning and limited experimentation scope.

Another challenge was the lack of detailed clinical metadata in the dataset, such as tumor grade or patient demographics. These could have enhanced model explainability and clinical utility.

Additionally, without integration of radiologist annotations or segmentation masks, the model was trained solely on whole MRI slices, potentially reducing diagnostic precision . Future extensions could include attention-based models or integration with clinical EHR data to improve interpretability and diagnostic value.

Risk Mitigation and Quality Evaluation

To mitigate risks such as session timeouts and data loss, we employed persistent storage and saved regular model checkpoints. Stratified train-validation splits were used to preserve label distribution and avoid bias. Overfitting was monitored through early validation, and evaluation relied on robust metrics like accuracy, F1-score, and confusion matrices to ensure reliability.

Academic and Professional Integration

The project followed real-world ML practices, including modular code design, version control, and documentation. Weekly mentor reviews mirrored agile sprint evaluations. Quality assurance was aligned with peer-reviewed standards and reproducibility principles. For future scaling, I plan to adopt tools like MLflow and align with ethical AI and clinical validation frameworks (Holzinger et al., 2022).

5.Level of the Project

Motivation and Real-World Relevance

The project aims to automate brain tumor classification from MRI scans using deep learning, a task with high clinical significance. Manual classification by radiologists is time-consuming and prone to subjectivity; automating this process can improve diagnostic speed and accuracy (Bharali et al., 2024). This aligns with my long-term goal of applying AI to critical healthcare problems, and reflects the current industry shift towards intelligent diagnostic systems.

Artefact Design and Problem-Solving Approach

The artefact is a deep learning pipeline that classifies brain MRIs into four categories: glioma, meningioma, pituitary tumor, and no tumor. I implemented a baseline CNN model Will implement ResNet and EfficientNet via transfer learning to benchmark improvements. Data augmentation and stratified splitting were used to address class imbalance and overfitting, contributing to model generalizability across tumor types - a necessity in healthcare AI systems (Castelino et al., 2024).

Research and Technical Complexity

The project integrates both theoretical research and practical execution. I conducted an in-depth literature review on CNN and transfer learning approaches specific to medical imaging, then translated those findings into the development and fine-tuning of architectures. Challenges such as GPU limitations were managed through batch optimization and resource planning, a common constraint in real-world AI development. This demonstrates the analytical and adaptive thinking required at an MSc level.

Testing, Evaluation and Robustness

Robust evaluation was conducted using stratified train-validation-test splits, along with accuracy, confusion matrix, and class-wise performance metrics. To better handle the clinical implications of false positives and false negatives, particularly in underrepresented tumor classes . This strategy ensures the model is both statistically valid and clinically applicable.

Tools, Methodologies and Justification

Industry-standard tools such as TensorFlow, Keras, Pandas, and Matplotlib were used for implementation, visualization, and experimentation. The project was managed using an Agile-inspired iterative cycle with weekly deliverables. This approach helped adjust priorities and improve system design over time, reflecting real-world engineering practices. The choice of transfer learning methods was validated both by literature and empirical performance, making them appropriate for limited yet high-dimensional medical data.

6.Conclusion

In this project, we developed a custom Convolutional Neural Network (CNN) model for classifying brain MRI scans into four categories: glioma, meningioma, pituitary tumor, and no tumor. Through targeted data balancing, extensive augmentation, and dropout regularization, the model achieved a strong test accuracy of 92% and high F1-scores, demonstrating its potential to support radiological diagnosis. This approach addresses the need for consistent and accurate brain tumor identification in clinical settings where manual interpretation is time-consuming and error-prone.

In the next phase, we will implement advanced transfer learning techniques using EfficientNet, ResNet, and Xception to improve accuracy and reduce training time. Additionally, ensemble learning methods will be explored to enhance prediction reliability and model generalization. These enhancements aim to create a scalable diagnostic tool that aligns with real-world medical workflows and supports radiologists in early and accurate brain tumor detection.

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